SPE-229665-MS

Co-Injection of Foam and Steam in SAGD Using a Modified Well Configuration-A Simulation Study

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This paper was prepared for presentation at the ADIPEC held in Abu Dhabi, UAE, 3 – 6 November 2025.

Abstract

This study investigates the technical viability of combining foam injection with steam in Steam-Assisted Gravity Drainage (SAGD) operations through modified well designs, employing both computational modeling and experimental analysis. The research utilizes a foam system comprising water, non-condensable gas, and surfactants, specifically designed to control gas mobility and minimize residual oil saturation. The foam creates an enhanced thermal barrier beneath the formation cap, effectively reducing heat transfer to overburden strata and improving cumulative Steam-to-Oil Ratio (cSOR).

The investigation employs vertical injection wells to position the foam directly under cap rock formations. Numerical simulations were conducted using CMG STARS thermal reservoir simulator, with Long Lake Pad 16 serving as the field prototype. All simulations incorporated no-flow boundaries, with producer wells constrained by maximum vapor injection rates and minimum bottomhole pressure thresholds.

A baseline scenario modeling conventional SAGD operations from 2017 to 2027 projected 75,000 m^3 of cumulative oil production with a cSOR of 6.19. Comparative analysis revealed that foam co-injection significantly enhanced both oil recovery and thermal efficiency. The foam mechanism effectively moderated gas mobility while reinforcing the insulating gas-saturation layer at the chamber apex, concurrently increasing trapped gas content and decreasing residual oil.

Optimal configurations demonstrated substantial cSOR improvements: a three-vertical-injector arrangement achieved 4.3 at 2,500 kPa, while a field-adaptable four-well system at 2,500 kPa (within Long Lake's 2,600 kPa limit) yielded a cSOR of 4.25. These findings confirm foam-assisted SAGD's potential to enhance heavy oil recovery while optimizing thermal efficiency through engineered well architectures.

Keywords: Foam, SAGD, Heat Loss, Oil Sands, Steam to Oil Ratio

Introduction

The province of Alberta, Canada, holds one of the world's largest unconventional hydrocarbon deposits in the form of heavy oil and bitumen (Dusseault, 2001). Due to the depth and geological nature of these reservoirs, surface mining is feasible for only a small portion, making in-situ thermal recovery methods essential for bitumen extraction. Among these, Steam-Assisted Gravity Drainage (SAGD) has emerged as a widely adopted approach due to its capacity to mobilize extremely viscous bitumen by injecting thermal energy in the form of steam.

In a conventional SAGD process, a pair of horizontal wells is placed with a vertical separation of 5 to 10 meters. Steam is injected into the upper well, creating a steam chamber that heats the surrounding bitumen. As heoilviscosity decreaseswith increasingtemperature (Fig. 1), gravity causes the mobilized oil to drain into the lower production well. While conceptually efficient, SAGD in field applications faces several technical and economic constraints that limit its recovery factor and increase operating costs.

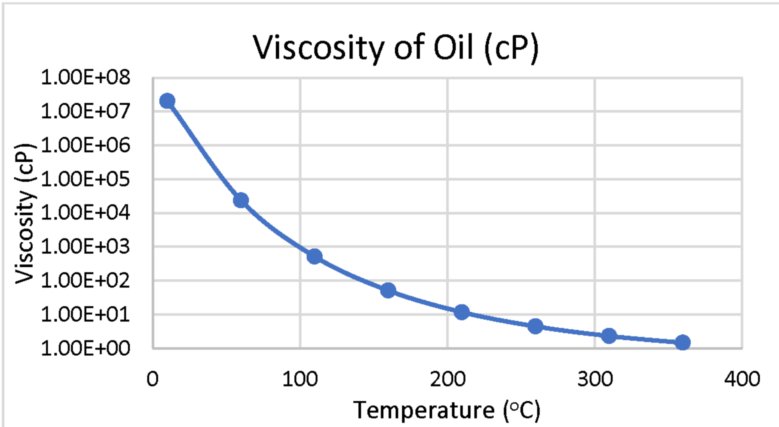

Figure 1—Temperature Dependence of Bitumen Density (Xu, 2015)

One primary concern is the significant heat loss from the steam chamber to the overburden and underburden formations. These thermal losses decrease the energy efficiency of the process and elevate the cumulative steam-oil ratio (cSOR), thereby increasing fuel and water consumption. In reservoirs with highvertical/horizontal permeability ratio or caprock proximity, the steam chamber may reach the top of the reservoir prematurely, exacerbating heat loss. Additionally, geological heterogeneities can cause uneven steam chamber development, resulting in poor sweep efficiency and early steam breakthrough.

Historical data from projects like Nexen's Long Lake operation show that high cSOR values, often exceeding 6, are associated with suboptimal chamber conformance and thermal losses. This reduces the economic viability of the process, particularly during periods of low oil prices.

To address these limitations, this study explores the co-injection of steam with two additives: non-condensable gas (NCG) and foam. NCG, such as methane, accumulates near the top of the chamber due to buoyancy and acts as an insulating gas cap, reducing vertical heat loss. Foam, formed in-situ by injecting surfactants with steam and gas, serves as a mobility control agent, diverting flow from high-permeability zones and enhancing steam distribution.

This paper evaluates a novel SAGD configuration using vertical steam injectors to more effectively place foam and NCG near the reservoir top. The schematic of the modified well configuration used in this study is shown in Fig. 2. Through thermal simulations using CMG STARS, we assess various scenarios in a Long Lake sector model and quantify improvements in cSOR, thermal efficiency, and oil recovery. The results demonstrate the potential of these enhancements to improve SAGD performance under realistic field conditions and regulatory constraints.

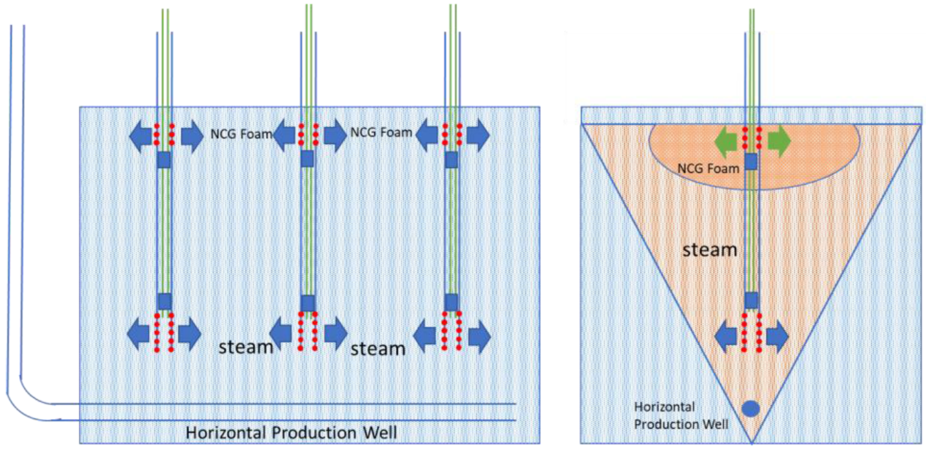

Figure 2—Schematic Diagram Foam Injection with A Modified Well Configuration (Zhang, 2024)

Review of Additive-Assisted Thermal Recovery in SAGD

The foundation of SAGD was established by Butler (1997), who proposed analytical models describing steam chamber development, gravity drainage, and heat transfer. While these models offer valuable theoretical insight, actual field operations often deviate due to factors such as reservoir heterogeneity, caprock geometry, and operational pressure limits. Vertical steam override and early breakthrough are especially common in layered or anisotropic formations, resulting in inefficient heat utilization and incomplete recovery.

To mitigate these issues, researchers have investigated the use of steam additives to improve both energy efficiency and sweep efficiency. Among these, non-condensable gases (NCGs) such as methane, nitrogen, and carbon dioxide have gained interest. Alturki et al. (2010) conducted simulation studies showing that NCGs can form a gas cap at the top of the chamber, acting as a thermal insulator and stabilizing the chamber shape. While promising, the effectiveness of NCGs is reduced in heterogeneous reservoirs where gas channeling or bypassing may occur, leading to reduced insulation and potential interference with steam propagation.

Foam, as an in-situ generated gas-liquid dispersion, offers an alternative means of mobility control. Hirasaki (1989) demonstrated that foam can reduce gas relative permeability, therebycontrolling gas override and promoting more uniform vertical and lateral sweep. Foam quality, which is determined by factors such as gas-liquid ratios, surfactant concentration, and reservoir conditions, directly affects its stability and flow resistance. High-quality foam increases apparent gas viscosity and blocks high-permeability channels.

Maini and Ma (1985) conducted extensive laboratory experiments to assess the stability of foam under SAGD-relevant conditions, including temperatures up to 300 °C and pressures above 5 MPa. They found that alkyl benzene sulfonates were particularly suitable due to their thermal and chemical resilience. Their findings also highlighted the importance of surfactant formulation in maintaining foam structure in the presence of oil, salinity, and temperature fluctuations.

Recent advances, including the work of Zhang (2018), have integrated both NCG and foam into SAGD models using alternative well configurations. By using vertical injectors to place the foam-gas mixture near the caprock, the insulating and sweep-enhancing benefits of both additives were enhanced. Simulation results showed that co-injection strategies could outperform either additive used alone in terms of cSOR reduction and thermal efficiency.

These studies collectively underscore the potential of additive-enhanced SAGD but also emphasize the importance of placement strategy, additive selection, and reservoir-specific calibration. The present study builds on this foundation by applying a combined foam-NCG approach to a realistic Long Lake reservoir model using vertical-well injectors to improve additive placement and overall thermal performance.

Methodology

This study employed thermal reservoir simulation using CMG STARS, a widely used tool for modeling steam-based recovery processes. The model is based on a representative sector of the Long Lake reservoir in Alberta, characterized by unconsolidated sandstone formations, high porosity (~30%), high permeability (1 to 3 Darcies), and extremely viscous bitumen with viscosities exceeding 100,000 cP at reservoir temperature.

The model domain includes a 25 m thick reservoir with laterally continuous sands, incorporating realistic geological heterogeneity derived from well logs, seismic interpretation, and core data. The grid comprises 60 × 50 × 20 cells in the I, J, and K directions, respectively, offering fine vertical and lateral resolution to capture steam chamber development and thermal fronts. No-flow boundary conditions were applied on all sides.

Initial reservoir conditions, including temperature, pressure, oil saturation, and water saturation, were calibrated using data from Nexen's field operations. Bitumen viscosity was modeled using a temperature-dependent correlation based on experimental data from Ashrafi et al. (2011), ranging from 300,000 cP at 20 °C to less than 5 cP at 300 °C. Methane properties such as K values (Fig. 3) were modeled using PVT correlations from Reid et al. (1977), and gas-liquid equilibrium was handled via CMG's thermal flash algorithm.

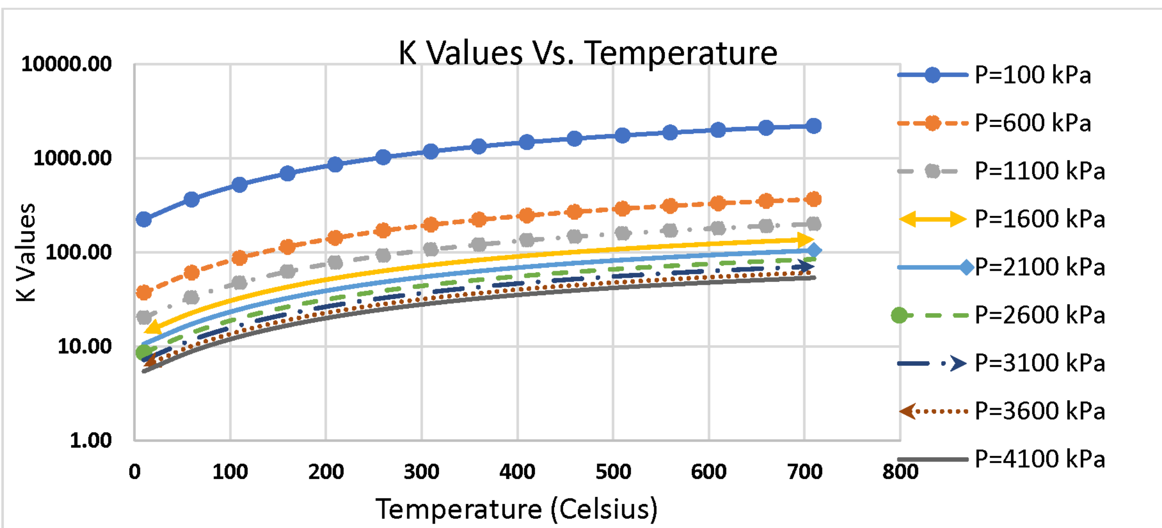

Figure 3—K Values of Methane at Different Temperature and Pressure (Reid, Prausnitz, & Sherwood, 1977; Zhang, 2024)

Foam Mechanism in SAGD

Foam, in the context of SAGD, functions as both a mobility control and thermal insulation agent (Hirasaki, 1989). It consists of gas bubbles encapsulated by surfactant-stabilized liquid films (lamellae). In porous media, foam is generated in situ through mechanisms such as leave-behind, lamella division, and snap-off, which occur when gas and surfactant-laden water are co-injected and interact with the pore structure. These mechanisms are triggered by shear forces, pore throat restrictions, and fluid saturation gradients.

In the steam-foam-enhanced SAGD process, the co-injection of surfactant, methane, and steam promotes foam generation within the reservoir, particularly near the injector. Once formed, foam significantly reduces gas mobility by increasing the apparent viscosity, especially in high-permeability regions and near the top of the reservoir where gas tends to accumulate. This is critical for mitigating steam override and channeling, which are common challenges in conventional SAGD.

However, the thermal conditions in SAGD reservoirs (up to 300 °C) pose challenges to foam stability. Surfactant selection is therefore crucial. Laboratory tests by Maini and Ma (1985) identified alkyl benzene sulfonates as suitable candidates due to their resilience under high temperature, salinity, and pressure. These surfactants maintain foam structure over extended periods and resist degradation in the presence of bitumen and divalent ions such as calcium and magnesium.

In this study, the foam system is designed to form near the vertical injectors, occupying the upper regions of the reservoir. By reducing gas mobility, foam traps methane near the caprock, thereby stabilizing the insulating gas cap and minimizing vertical heat losses. Simultaneously, it restricts flow in thief zones, promoting lateral chamber growth and improved contact with lower-permeability regions. As a result, foam enhances thermal efficiency and improves steam conformance.

The foam model in CMG STARS(Computer Modelling Group Ltd., 2017)is empirical and modifies gas relative permeability using a foam mobility reduction factor (FM)(Equation 1), which is a function of local surfactant concentration, oil saturation, and capillary number.

$$FM = \frac{1}{1 + FMMOB^{*} \quad F1^{*} \quad F2^{*} \quad F3^{*} \quad F4^{*} \quad F5^{*} \quad F6} \qquad \text{Equation 1}$$

Where,

$$F1 = \left[\frac{\text{Mole Fraction (ICPREL)}}{\text{FMSURF}}\right]^{EPSURF} \qquad \text{Equation 2}$$

$$F2 = \left[\frac{FMOIL - S_o}{FMOIL - FLOIL}\right]^{EPOIL} \qquad \text{Equation 3}$$

$$F3 = \left[\frac{FMCAP}{\text{Capillary Number}}\right]^{EPCAP} \qquad \text{Equation 4}$$

$$F4 = \left[\frac{FMGCP - \text{ Capillary Number}}{FMSURF}\right]^{EPGCP} \qquad \text{Equation 5}$$

$$F5 = \left[\frac{FMOMF - \text{ Oil Mole Fraction (x)}}{FMOMF}\right]^{EPOMF} \qquad \text{Equation 6}$$

$$F6 = \left[\frac{\text{MOLE FRACTION (NUMW)} - FLSALT}{FMSALT - FLSALT}\right]^{EPSALT} \qquad \text{Equation 7}$$

A combination of six foam sensitivity functions (F1 to F6) accounts for these parameters (Equation 1 to Equation 7). In this study, the following key values were used:

- Critical surfactant mole fraction: 1.875×10^{-4}
- Critical oil saturation: 0.5
- Maximum foam strength (FMMOB): 50
- krg base values: ranged from 0.05 to 0.2 in sensitivity analyses

The foam was modeled as forming in-situ from the simultaneous injection of steam, methane (as NCG), and a thermally stable surfactant. The injection compositions and mass fractions were carefully controlled to remain within the foam generation envelope. The goal was to simulate conditions where foam forms near the injector and propagates laterally and vertically, providing mobility control and thermal insulation.More specifically, the foam model in this study simulates foam strength through a mobility reduction factor (FM), where lower FM indicates stronger foam. The reference factor (FMMOB) is set to 50. Foam strength depends on surfactant concentration (F1), oil saturation (F2), and capillary number (F3). As surfactant concentration increases, foam strength rises and peaks at a critical mole fraction of 1.875E-4. Foam collapses when oil saturation exceeds 0.5 or when the capillary number surpasses 1E-4. The sensitivity of foam strength to these variables is controlled by exponents (EPSURF, EPOIL, EPCAP), where lower exponent values intensify foam response. In this study, the effects of generation capillary number, oil mole fraction, and salt mole fraction (F4–F6) were considered negligible compared to surfactant mole fraction, oil saturation, and capillary number. As a result, default values of 1.0 were assigned to F4–F6, indicating no contribution to foam behavior.

Simulation Setup

Model Dimension. A detailed geological model of the Long Lake reservoir (Fig. 4), provided by Nexen Inc., was used in this study, focusing on Pad 16. The selected region spans 590 m × 50 m × 55 m, with fine grid resolution and over 71,000 active blocks. The model features realistic top and bottom boundary curvatures and includes a standard SAGD well pair spaced 5 m apart, with the producer positioned 1 to 2 m above the bottom water zone to prevent steam breakthrough.

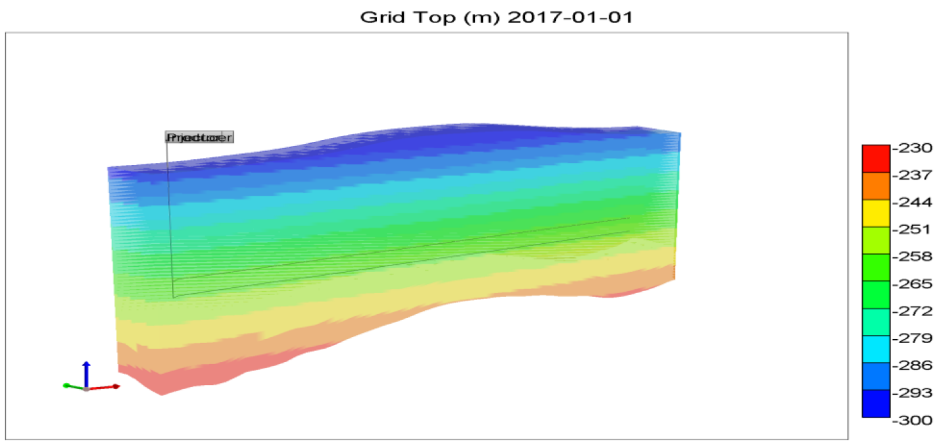

Figure 4—3D View of Simulation Model (Zhang, 2018)

Geological Model. The targeted reservoir zone at Long Lake is highly heterogeneous, with significant spatial variation in porosity, permeability, saturation, and net-to-gross (NTG) ratio. NTG values are close to one in oil zones but drop to zero in the bottom water zone and interbedded layers. Horizontal permeability averages 3,796.69 mD, mostly between 2,000 and 6,000 mD, while vertical permeability averages 1,444.54 mD, with over half of the blocks below 1,000 mD. Porosity ranges from 0.0057 to 0.395 (average 0.29), and oil saturation is significantly higher in the oil zone, averaging 0.576, compared to water saturation, which averages 0.424 and is higher in the bottom water zone.Key reservoir characteristics in the vertical section of the geological model are presented in Fig. 5.

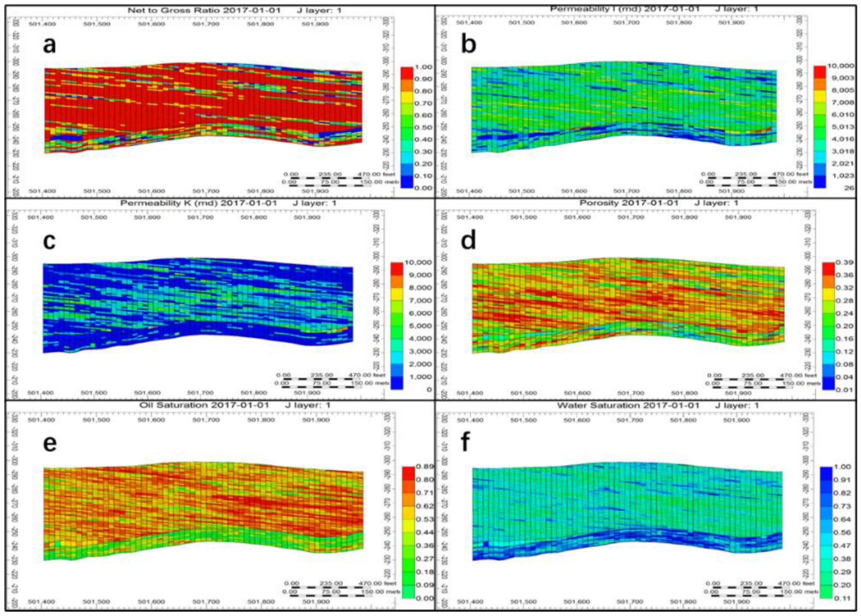

Figure 5—Geological Model ((a-f) shows Net to Gross Ratio, Horizontal Permeability, Vertical Permeability, Porosity, Oil Saturation, and Water Saturation)

Simulation Scenarios. To assess the effectiveness of foam and NCG co-injection under different operational settings, six primary simulation scenarios were modeled. These scenarios were designed to isolate the effects of individual additives and explore the influence of injection configuration and pressure. Performance metrics such as cumulative oil production, cSOR, oil recovery factor, and steam chamber growth were recorded for all scenarios. These results are used in the following sections to assess the technical and economic viability of the proposed enhancements.

Base Case

Conventional SAGD using a horizontal injector and producer pair with steam-only injection. This case serves as a performance baseline. In the base case scenario, a conventional SAGD well pair is used without any steam additives, with operations running from January 2017 to January 2027. A three-month preheating period is simulated using heater wells to establish communication between the injector and producer. Reservoir and fluid properties are derived from Nexen's geological model. Relative permeability curves are constructed based on Xu (2015) to match field conditions, targeting the observedsteam-oil ratio (SOR) of 6 to 7.Initial reservoir temperature and pressure are set at 12 °C and 1,100 kPa, respectively. The simulation uses a surface injection steam temperature of 208 °C with 95% quality and a producer bottom-hole pressure limit of 1,100 kPa. Due to low injectivity in vertical wells under low pressure, an additional high-pressure base case (3,500 kPa) is established to provide a valid performance benchmark for comparison with modified well configurations.

NCG-Only Injection using Conventional Well Configuration

To evaluate the impact of non-condensable gas (NCG) on SAGD performance, a series of simulations were conducted using conventional well configurations at a low operating pressure of 1,867 kPa. Steam and NCG (methane) were co-injected at various mole fractions ranging from 0.25% to 1.5%. The surface fluid rate was constrained instead of the steam rate, with total injection volumes adjusted according to the NCG concentration. A maximum steam rate of 300 m^3/day was maintained until November 2020, after which NCG injection began, coinciding with the steam chamber reaching the top of formation. This setup served as a reference forassessing the effectiveness of NCG in improving oil recovery and reducing the steam-oil ratio (SOR) prior to testing modified well configurations.

Steam-Only Injection using Modified Well Configuration

This case investigates a modified SAGD configuration using only vertical steam injectors positioned above a horizontal producer, aiming to delay steam chamber contact with the reservoir top and reduce overburden heat loss. Inspired by early SAGD developments and field success at the Sceptre project, steam is injected through multiple vertical wells spaced 3 m above the producer, promoting the formation of separate steam chambers that later merge. However, simulations at 1,867 kPa bottom-hole pressure revealed low steam injectivity and limited oil production. To overcome this, a higher injection pressure of 3,500 kPa was tested. The original horizontal injector is replaced by three or four vertical injectors, with perforation count increased from 5 to up to 15 per well to enhance injectivity. This configuration aims to improve thermal efficiency while addressing early steam breakthrough and injectivity limitations.

NCG-Only Injection using Modified Well Configuration

In this case, a modified well configuration is employed where steam and methane (NCG) are co-injected through the same vertical wellbore but at different depths, maintaining at least 5 m of vertical separation. Methane is injected near the top of the formation through upper perforations to form a cold gas layer that insulates the steam chamber and reduces heat loss to the overburden. Steam is injected from lower perforations using separate well completions. NCG mole fraction sensitivity cases ranging from 0.25% to 1.5% were simulated by assigning separate injectors for steam and gas, with appropriate flow constraints. Packers are assumed to prevent mixing within the wellbore. To ensure regulatory compliance, an additional simulation was conducted with a maximum bottom-hole pressure (BHP) of 2,500 kPa, which reflects the field-approved pressure limit at Long Lake. Performance at this pressure was compared using 3- and 4-well injector configurations.

Foam-NCG Co-Injection (3 injectors)

In the foam injection case, a modified well configuration is used to co-inject methane, surfactant, and water at a pressure of 1,867 kPa and 20 °C, forming a cold foam system with 70% reservoir foam quality.Surface condition foam quality reaches approximately 97.8% due to methane expansion. The surfactant concentration is set at 0.1 vol%, and gas-to-liquid ratios are adjusted to match those in the NCG-only cases for consistency. The data relevant to foam calculations are shown in Table 1. Sensitivity cases are conducted for methane mole fractions ranging from 0.25% to 1.5% using both three- and four-well configurations. The corresponding foam volume analysis for the three-well scenario is provided inTable 2. Foam behavior is simulated using CMG STARS' empirical model, which reduces gas mobility by modifying the gas relative permeability curve. Endpoint krg values are varied between 0.005 and 0.03 to assess the impact of foam strength on reservoir performance. These cases evaluate foam's effectiveness in enhancing thermal insulation, reducing gas mobility, and improving steam conformance.

Table 1—Foam Calculation Results in Foam Injection Case

Based on 1 m^3	Volume injected	1	m^3
	Concentration	0.1	%
Foam Quality Reservoir Condition	Water and Surfactant	0.3	RC
	Methane	0.7	RC
Foam Quality Standard Condition	Water and Surfactant	0.3	SC
	Methane	13.13	SC
Standard Condition Volumes	Water and Surfactant	0.3	m^3
	Methane	13.13	m^3
	Surfactant	0.0003	m^3
	Water	0.2997	m^3

Volume Fractions Specified in Simulation	Methane	0.978	
	Surfactant	2.23E-05	
	Water	0.0223	

Table 2—Foam Volume Analysis at Foam Case with Three Modified Wells (Zhang, 2018)

Mole fraction (%)	NCG to Steam Ratio in NCG Case	Foam Volume (m³) at One Well
NCG In total Fluid (NCG and Steam Foam)	x	3 Well Case
1.5	19.89	2034.34
1.25	16.57	1695.29
1	13.26	1356.23
0.75	9.94	1017.17
0.5	6.63	678.12
0.25	3.31	339.06

Foam-NCG Co-Injection (4 injectors)

In the four-well foam injection case, the total methane volume is held constant across all scenarios. For methane mole fractions ranging from 0.25% to 1.5%, the required surface-condition fluid injection volume per well varies from 254.29 m^3 to 2,034.35 m^3, respectively. Foam composition follows the same volume fraction proportions as defined in the base calculation (Table1), with detailed injection volumes summarized in Table 3.

Table 3—NCG Volume Analysis at Foam Case with Four Modified Wells (Zhang, 2018)

Mole Fraction(%)	NCG to Steam Ratio in NCG Case	Foam Volume (m³) at One Well
NCG In total Fluid (NCG and Steam Foam)	x	4 Well Case
1.5	19.89	1525.76
1.25	16.57	1271.47
1	13.3	1017.17
0.75	9.95	762.89
0.5	6.63	508.59
0.25	3.31	254.29

Foam-NCG Co-Injection using Conventional Well Configuration

These cases evaluate foam's effectiveness in enhancing thermal insulation, reducing gas mobility, and improving steam conformance. In the conventional SAGD with foam case, hot NCG is combined with steam, surfactant, and water to form steam foam, as cold NCG cannot be injected alongside high-temperature steam in this configuration. A maximum steam rate of 300 m^3/day is maintained, and the injected fluid is defined as a WATER-GAS mixture in the simulation. Due to the higher liquid water content associated with steam, the surfactant volume is increased accordingly. The primary operational constraint is the total injected fluid volume, with a secondary constraint of 2,500 kPa maximum bottom-hole pressure. Foam composition varies with NCG mole fraction, with detailed component breakdowns provided in Table 4 and Table 5.

Table 4—Volume Fraction of Components in Foam (Zhang, 2018)

	0.25%	0.5%	0.75%	1.25%	1.5%
	Volume Fraction	Volume Fraction	Volume Fraction	Volume Fraction	Volume Fraction
Water	0.29	0.15	0.11	0.077	0.069
Surfactant	0.00025	0.00015	0.00011	7.79E-05	6.9E-05
Methane	0.75	0.85	0.89	0.92	0.93

Table 5—Total Fluid Volume Table at Different Mole Fraction of NCG (Zhang, 2018)

Steam (m^3)	NCG Mole Fraction (%)	Water in Foam (m^3)	Surfactant (m^3)	Methane (m^3)	Total Fluid Volume (m^3)
300	1.5	133.48	4.34E-01	5849.53	5.98E+03
300	1.25	111.24	4.12E-01	4874.61	4.99E+03
300	0.75	66.74	0.37	2924.77	2.99E+03
300	0.5	44.49	0.34	1949.84	1.99E+03
300	0.25	22.25	0.32	974.92	9.97E+02

Impact of Foam-NCG Interaction on SAGD Efficiency

The synergy between foam and NCG in SAGD provides significant benefits that neither additive can fully achieve alone. NCG, such as methane, is lighter than steam and does not condense. When injected, it naturally migrates to the top of the steam chamber, forming a gas cap that acts as a thermal blanket. This reduces heat conduction to the overburden, preserving the temperature within the chamber and enhancing the overall energy efficiency of the process.

However, in the absence of mobility control, NCG may bypass low-permeability regions or escape through high-permeability streaks, leading to early gas breakthrough and poor sweep. This limits its insulating capacity and may interfere with steam propagation. Foam mitigates this by reducing gas mobility, stabilizing the gas cap, and slowing vertical gas migration. The result is a more persistent insulating layer that maintains its position and effectiveness over time.

Foam also plays a critical role in improving steam distribution. By increasing gas phase resistance, foam diverts steam away from dominant flow channels into unswept or poorly contacted zones. This enhances vertical and lateral sweep efficiency, resulting in lower residual oil saturation and more uniform thermal fronts.

Simulation results from this study confirm these synergistic effects. Scenarios with NCG alone show moderate improvement in cSOR and temperature retention near the caprock. However, when foam is co-injected, thermal losses are further reduced, and steam breakthrough is delayed. The foam-stabilized gas cap leads to more consistent pressure and temperature profiles, greater cumulative oil production, and a lower cSOR.

These benefits are especially pronounced in heterogeneous reservoirs where conformance issues are more severe. Foam ensures that both steam and gas additives interact more efficiently with the reservoir matrix, maximizing heat transfer and minimizing bypassed oil. This translates into better overall process performance, both thermally and hydraulically.

From a design perspective, co-injection of foam and NCG also provides flexibility. Operators can adjust surfactant concentration, gas fraction, and injection timing to optimize performance under different reservoir conditions. For instance, foam strength can be tuned via krg values to balance injectivity and conformance, while alternating injection cycles (e.g., foam slugs) can prevent wellbore plugging or excessive pressure buildup.

In summary, the foam-NCG interaction offers a powerful tool for enhancing SAGD performance. It addresses key inefficiencies, such as heat loss and poor sweep, while remaining adaptable to operational constraints.These characteristics make the approach particularly attractive for mature or marginal SAGD projects where maximizing energy efficiency and recovery is essential.

Results and Discussion

Results Overview

A comparative analysis of six SAGD scenarios was conducted from January 2017 to January 2027 (Fig. 6). Among all cases, foam injection using the modified well configuration achieved the highest cumulative oil recovery at 103,199 m^3, while the conventional SAGD base case yielded the lowest at 75,199 m^3. The remaining four cases produced between 79,377 m^3 and 91,223 m^3. Notably, modified well configurations exhibited higher oil rates after 2024 due to the delayed but eventual coalescence of steam chambers.

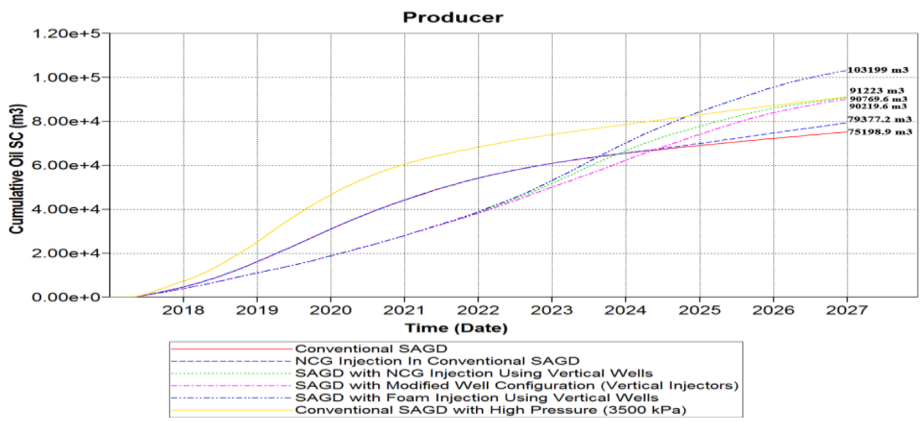

Figure 6—Cumulative Oil Rate of Six Scenarios (Zhang, 2018)

Cumulative steam-oil ratio (cSOR) trends (Fig. 7) show that all cases experienced a cSOR decline from 2017 to 2022, attributed to high initial steam injection for preheating. Foam-assisted SAGD with modified wells achieved the lowest cSOR at 4.31, followed by NCG-assisted SAGD at 4.65. In contrast, the high-pressure conventional SAGD case had the highest cSOR of 6.84, with the base case slightly lower at 6.19.

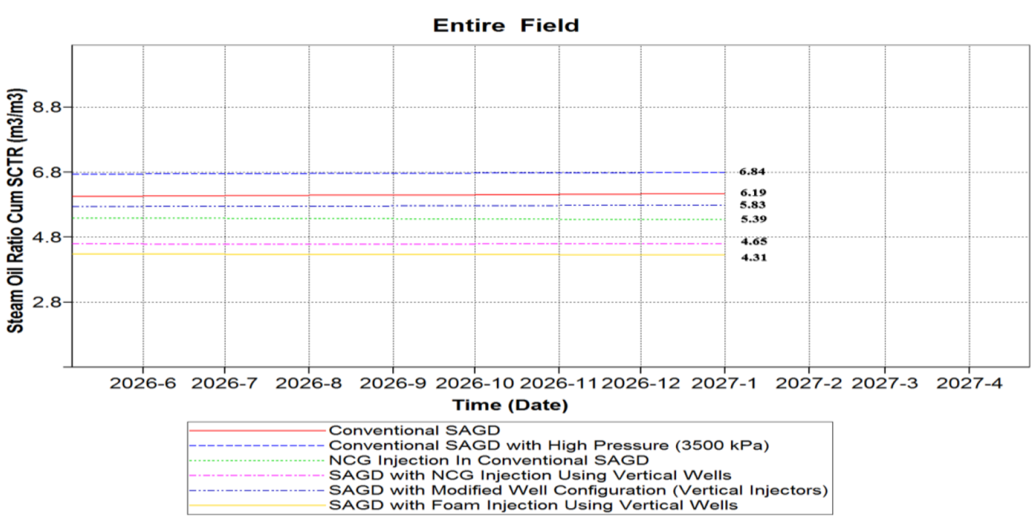

Figure 7—Detailed Cumulative Steam Oil Ratios of Six Scenarios (Zhang, 2018)

Steam chamber development (Fig. 8) revealed that NCG or foam should be injected after steam has reached the top of the reservoir to maximize efficiency. Early injection leads to premature thermal isolation and lower productivity. At the end of the simulation (Fig. 9), NCG and foam cases showed reduced temperatures near the caprock (from over 200 °C to ~150 °C), effectively limiting heat loss to the overburden.

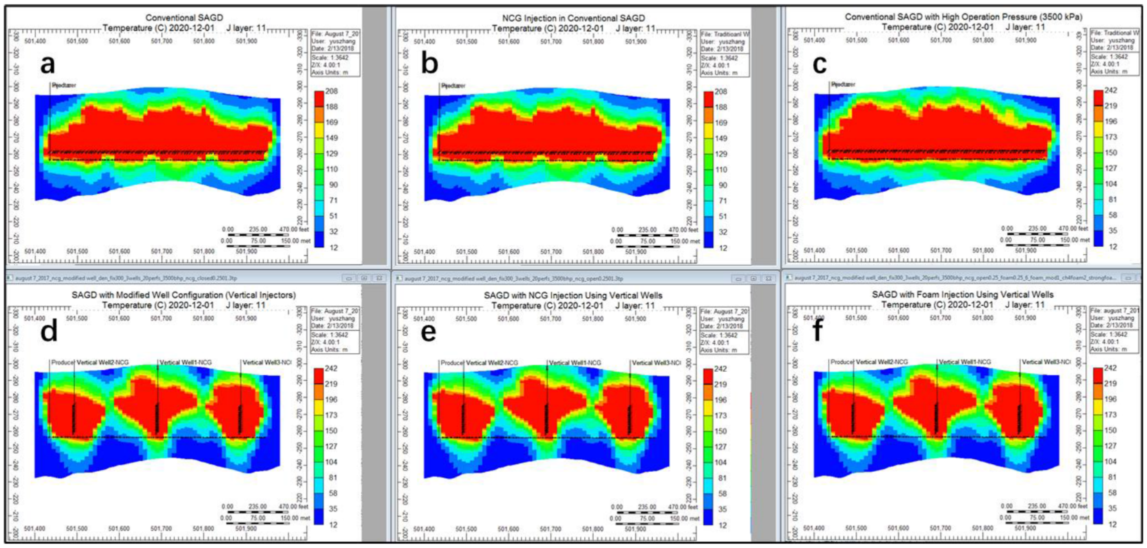

Figure 8—Steam Chamber Profile before Injecting NCG or Foam of Six Scenarios (Zhang, 2018)((a-f) shows Conventional SAGD, Conventional NCG-SAGD, Conventional SAGD with 3500kPa, SAGD using Modified Well Configuration, NCG-SAGD using Modified Well Configuration, and Foam-SAGD using Modified Well Configuration)

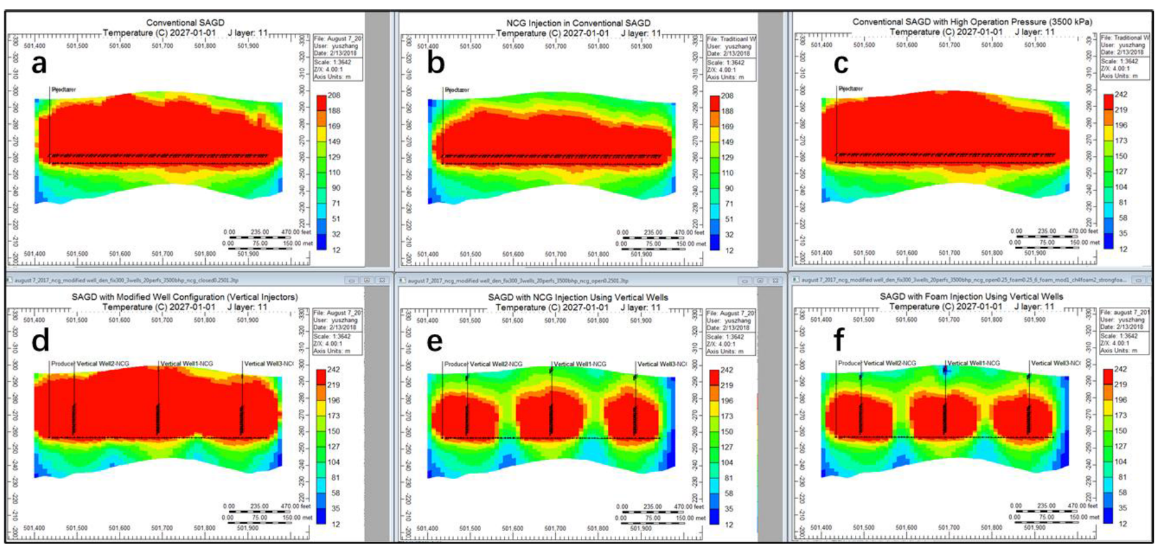

Figure 9—Steam Chamber profiles at the End of Simulation of All Six Scenarios (Zhang, 2018)((a-f) shows Conventional SAGD, Conventional NCG-SAGD, Conventional SAGD with 3500kPa, SAGD using Modified Well Configuration, NCG-SAGD using Modified Well Configuration, and Foam-SAGD using Modified Well Configuration)

Discussion

The simulation results clearly demonstrate the advantages of incorporating NCG and foam into the SAGD process. Each scenario was evaluated based on key performance metrics, including cumulative oil production, cSOR, temperature retention, steam chamber growth, and oil saturation distribution.

In the base case, a conventional SAGD operation using a horizontal well pair and steam-only injection was simulated. The model produced approximately 75,000 m^3 of oil over 10 years, with a cumulative steam-oil ratio (cSOR) of 6.19.This high cSOR is consistent with historical Long Lake field data and reflects significant thermal losses to the overburden. Thermal maps showed pronounced upward heat dissipation early in the steam chamber development, while oil saturation plots indicated limited sweep near the top of the reservoir.

Under pressure-constrained conditions at 2,500 kPa, the configuration was modified to include four vertical injectors to preserve injectivity and chamber growth. Even at this lower pressure, the steam and NCG scenario achieved a cSOR of 4.25, suggesting that the insulating effect of methane remains effective under regulatory pressure limits.

The most promising results were observed in foam-assisted SAGD scenarios. At 3,500 kPa with three vertical injectors, foam co-injection yielded the highest cumulative oil recovery (over 100,000 m^3) while achieving a relatively low cSOR of 4.31. Foam enhanced steam conformance by suppressing vertical steam override and encouraging lateral steam chamber growth. Isothermal profiles showed more uniform heat distribution, and gas saturation maps confirmed a stable foam layer beneath the caprock.

While the four-injector configuration at 3,500 kPa achieved a slightly higher oil recovery and marginally lower cSOR, the added cost of drilling a fourth vertical well reduces its practical economic advantage. Steam-foam injection using hot foam with 1.5 mol% NCG and a conventional well configuration achieved the lowest cSOR among all cases. However, it resulted in significantly lower oil recovery and increased consumption of NCG and surfactant, raising concerns over efficiency and chemical costs.

At 2,500 kPa with four vertical injectors, foam still performed effectively, yielding a cSOR of 4.25. Although slightly higher than the high-pressure case, the improvement over base SAGD (6.19) remained substantial. Foam helped maintain a consistent steam chamber profile with delayed breakthrough and lower residual oil saturation.

Sensitivity analyses revealed several key trends:

- Lower krg values (e.g., 0.1) produced stronger foam, improving cSOR, but very strong foam could reduce injectivity near the wellbore.
- Surfactant concentrations above the Critical Micelle Concentration (CMC) enhanced foam propagation but needed to be balanced with chemical cost and adsorption effects.
- NCG mole fractions of 0.25 to 0.4 provided optimal insulation without excessive gas override.

These results reinforce that foam and NCG not only improve heat retention but also reshape fluid flow in the reservoir. The combination reduces steam channeling, enhances contact with lower-permeability zones, and increases the uniformity of oil displacement. Performance is summarized as follows in Table 6:

Table 6—Summary of Simulation Results (Zhang, 2018)

Injected Fluid	Well Type for Injection	Injector BHP Constraint(kPa)	Cumulative Oil Production (m^3)	Cumulative SOR
Steam only	Horizontal well	1,867	75,200	6.19
Steam only	Horizontal well	3,500	91,223	6.84
Steam with 0.25 mole% NCG	Horizontal well	1,867	79,500	5.39
Steam only	3 Vertical wells	3,500	90,219	5.83
Steam & 0.25 mole% NCG (segregated injection method)	3 Vertical wells	3,500	90,800	4.65
Steam & 0.25 mole% NCG (segregated injection method)	3 Vertical wells	2,500	65,800	4.94
Steam & 0.25 mole% NCG (segregated injection method)	4 Vertical wells	2,500	70,700	4.63

Injected Fluid	Well Type for Injection	Injector BHP Constraint(kPa)	Cumulative Oil Production (m^3)	Cumulative SOR
Steam & foam (segregated injection with 0.25 mole% gas)	3 Vertical wells	3,500	103,200	4.31
Steam & foam (segregated injection with 0.25 mole% gas)	4 Vertical wells	3,500	108,000	4.23
Steam & foam (segregated injection with 0.25 mole% gas)	3 Vertical wells	2,500	72,000	4.72
Steam & foam (segregated injection with 0.25 mole% gas)	4 Vertical wells	2,500	80,400	4.25
Steam with hot foam using 1.5 mole% gas	Horizontal well	1,867	92,900	4.02

These improvements suggest that foam and NCG co-injection is a technically robust enhancement that delivers both recovery and energy efficiency benefits under a range of operating conditions.

Operational Considerations and Field Implementation

Successful field deployment of foam and NCG co-injection requires careful integration of chemical, mechanical, and operational systems. The first consideration is surfactant selection. As demonstrated in laboratory studies, alkyl benzene sulfonates are ideal for SAGD due to their high thermal stability, low degradation rates at 250 to 300 °C, and good compatibility with formation brine.Surfactant choice must also consider adsorption losses on reservoir rock.

Injection strategy plays a key role in operational success. While continuous co-injection of steam, gas, and surfactant is simple in design, it can lead to foam accumulation near the wellbore, increasing pressure and risking formation damage. To mitigate this, a slug-based injection method is often preferred, where foam slugs alternate with steam-only periods. This allows for pressure relief and better foam propagation into the reservoir.

Surface facility upgrades are necessary for surfactant storage, mixing, and dosing. These include heated tanks, chemical injection pumps, inline mixers, and real-time control systems. It is also critical to isolate surfactant lines from direct contact with high-temperature steam until injection, to prevent premature degradation.

Monitoring and diagnostics such as fiber-optic distributed temperature sensing (DTS), pressure gauges, and gas/oil ratio tracking are essential. These tools enable operators to assess steam chamber growth, detect early gas breakthrough, and evaluate foam effectiveness in situ. Tracer tests may also be employed to study foam propagation and validate model predictions.

Finally, injectivity management is essential. Foam's high apparent viscosity may reduce injectivity, especially during initial propagation. Pressure must be monitored closely to avoid exceeding caprock integrity limits. Pilot-scale operations are recommended before full-field deployment to fine-tune injection rates, chemical doses, and pressure management protocols.

Economic and Environmental Implications

The integration of foam and NCG into SAGD operations provides not only technical benefits but also compelling economic and environmental advantages. The most direct economic impact stems from the significant reduction in steam requirements. In SAGD, steam generation can account for over 50% of operating costs due to high fuel consumption, water treatment, and boiler maintenance. Reducing the cSOR from 6.19 to 4.3, as achieved in the optimized foam scenario, represents a nearly 31% reduction in steam usage. This reduction results in substantial cost savings over the project life cycle.These savings are even more pronounced under high natural gas prices or carbon pricing schemes. In Alberta, SAGD operators are subject to emissions intensity benchmarks and carbon taxes. Lower steam usage translates to lower fuel combustion and reduced CO_2 emissions, helping operators meet regulatory targets and improve their ESG performance metrics. In emissions-intensive processes like SAGD, such reductions have direct financial implications, including lower compliance costs and carbon credit opportunities.

In terms of project longevity, reduced thermal input and better conformance extend the productive life of SAGD well pairs. Higher oil recovery from existing infrastructure minimizes the need for new drilling, reducing capital expenditure and environmental disturbance. Furthermore, less water is required for steam generation, easing demands on freshwater resources and water treatment facilities.

From an environmental perspective, improved thermal efficiency directly reduces greenhouse gas emissions per barrel produced. For each unit of steam saved, both fuel demand and water consumption are reduced, decreasing the overall environmental footprint of the operation. This aligns with growing stakeholder expectations and government mandates for cleaner oil sands production.

In summary, the co-injection of foam and NCG represents a practical, near-term method to improve SAGD economics while simultaneously supporting emissions reduction and sustainability goals.

Conclusions

This study presents a comprehensive simulation-based evaluation of steam-foam and non-condensable gas co-injection in SAGD, using modified vertical injector configurations. The key findings are:

1. Foam and NCG co-injection reduced cSOR by up to 32% compared to base SAGD, improving thermal efficiency and oil recovery.
2. Methane formed a stable gas cap, reducing vertical heat loss and enhancing chamber insulation.
3. Foam reduced gas mobility, stabilized the gas cap, and improved steam conformance, especially in heterogeneous formations.
4. Vertical injectors allowed better additive placement, optimizing steam chamber geometry and contact with bitumen-rich zones.
5. Surfactant stability and foam strength (e.g., krg= 0.1) were critical to balancing conformance and injectivity.

These results confirm the technical potential of foam-NCG enhanced SAGD as a feasible, cost-effective, and environmentally beneficial method, especially in mature or marginal bitumen reservoirs.

Recommendations and Future Work

While simulation results are promising, field-scale implementation requires careful piloting and further research. A dedicated pilot test is recommended to validate the foam-NCG co-injection strategy under real reservoir conditions. The pilot should include:

- Vertical steam injectors positioned near the caprock.
- Real-time temperature and pressure monitoring using fiber optics.
- Surfactant and gas breakthrough tracking via chemical tracers.

Pilot results will provide essential data for optimizing injection schedules, foam slug size, and surfactant concentration profiles. They will also help assess surfactant retention, degradation rates, and injectivity limitations.

Further laboratory studies are needed to tailor surfactant formulations to Long Lake reservoir conditions. Factors such as salinity, divalent ion content, and temperature must be considered to avoid surfactant precipitation or foam collapse. Stability tests at 250 to 300 °C using actual reservoir brine samples are essential for surfactant screening.

Numerical models can also be improved by incorporating temperature-dependent foam degradation kinetics, adsorption isotherms, and rock-fluid interaction effects. Including geomechanical coupling in simulations would allow for assessment of caprock integrity, especially under high foam pressure buildup scenarios.

On a broader scale, full-field economic modeling should be conducted, incorporating capital and operating costs for surfactant injection, chemical storage, monitoring equipment, and incremental recovery. These cost models can be integrated with life-cycle emissions assessments to quantify the long-term value of foam-NCG SAGD enhancements.

Acknowledgement

I would like to express my sincere gratitude to CNOOC International, the Natural Sciences and Engineering Research Council of Canada (NSERC), and the University of Calgary for their generous financial support during my Master's and Ph.D. studies.

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